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Group No: 1

1. Structural analysis

Assumptions

Assumed lengths:

AB = 250mm, BE = 1050mm, CD = 250mm,
BC = 540mm, BS_2 = 320mm, BD = 592.5mm.

A 45° angle is assumed for link 2 with respect to the horizontal and CD ⊥ BE.
The drawing was done in 1:2 scale.

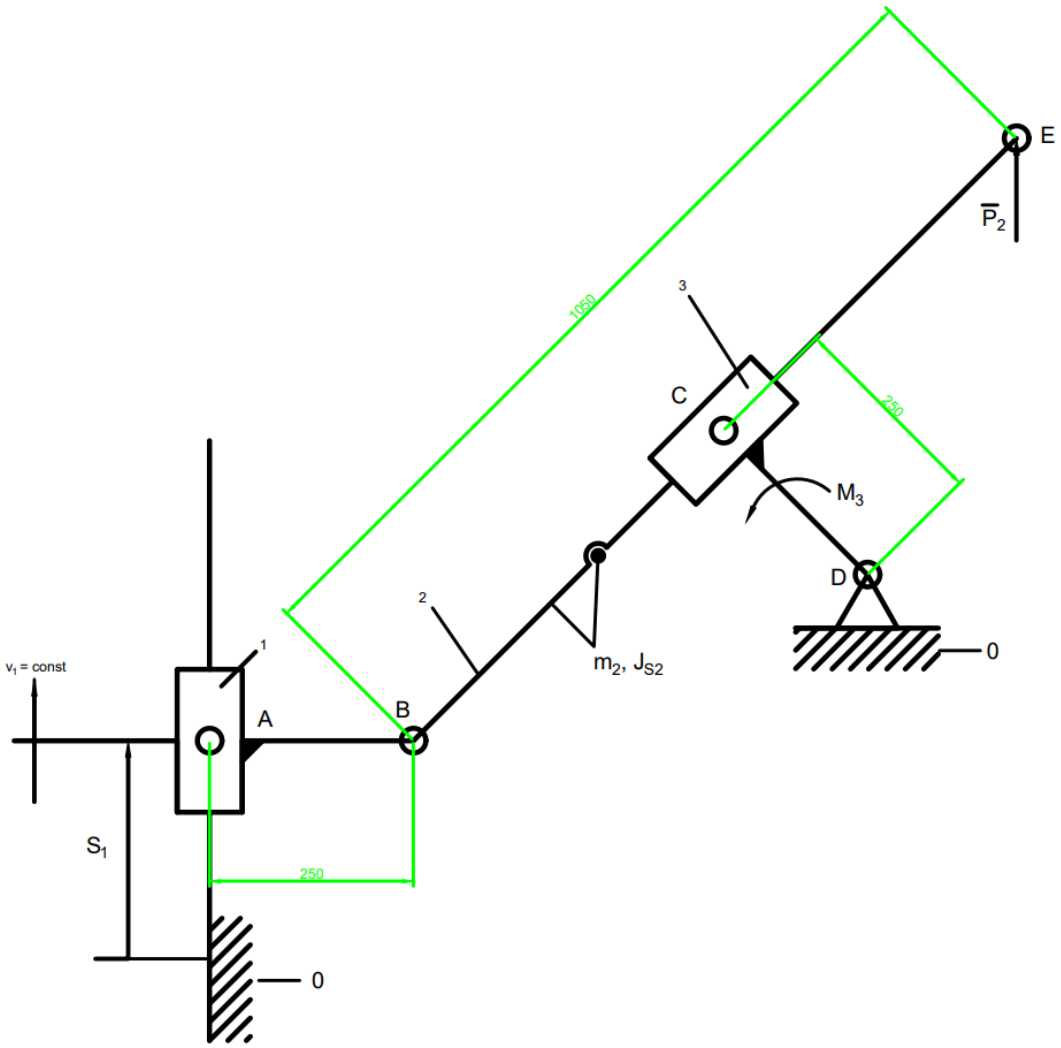

Fig. 1 - The mechanism with assumed lengths.

Mobility

In order to calculate the mechanism's mobility, i.e. the total amount of its degrees of freedom, the following formula can be used:

$$w = 3n - 2p_5 - p_4,$$

where:

w - mobility;

n - number of moveable links in the kinematic chain;

p_5, p_4 - number of 5th and 4th class of kinematic pairs.

The above formula is a simplified form used for plane mechanisms so it is applicable in this case. The number of moveable links in this case is 3. There are four kinematic pairs:

- **0-1:** class V as the slider can only move across the Y axis,
- **1-2:** class V as there is only rotational motion about point B,
- **2-3:** class V as the slider can only move across the beam BE,
- **3-0:** class V as the link can only rotate about point D.

Thus, there are four pairs of class V and none of class IV:

$$w = 3 \cdot 3 - 2 \cdot 4 - 0 = 1.$$

The mobility is 1, meaning there is only one DOF, so only one motor would be necessary to operate the positions of all the system's elements.

After designating link 1 as the driver link, there are two remaining links with three kinematic pairs of class V - thus, the mechanism is of class 2.

The form of the structural group is: – T – (R – T – Rs –), where:

T - translation;

R - rotation;

Rs - rotation of the slider.

This is a form 3 structural group of class 2.

2. Kinematic analysis

Graphoanalytical approach

Assumptions:

$v_1 = 0.5\frac{m}{s}$, $s_1 = 250mm$, scale: $10\ mm = 0.1\frac{m}{s}$.

Having assumed v_1 as the linear velocity of link 1 (i.e. of point A), it can be seen it will also be the velocity of point B, as they are connected:

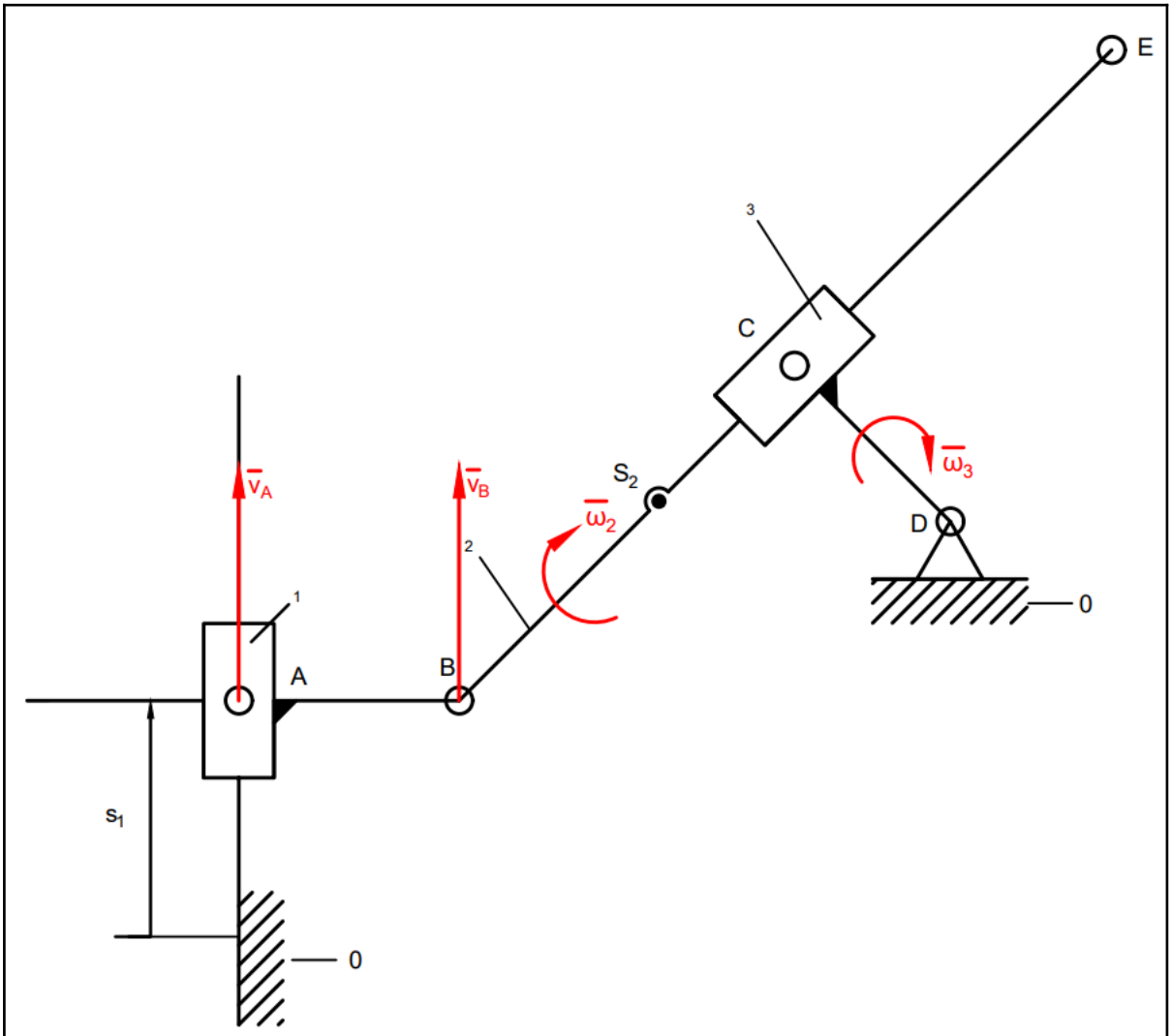

Fig. 2 - Known linear velocities v_A, v_B, and angular velocities ω_2 and ω_3 (with assumed directions).

Since the two angular velocities have the common point C, they will have the same value. To find this value, an equivalent system has to be assumed as seen in Fig. 3.

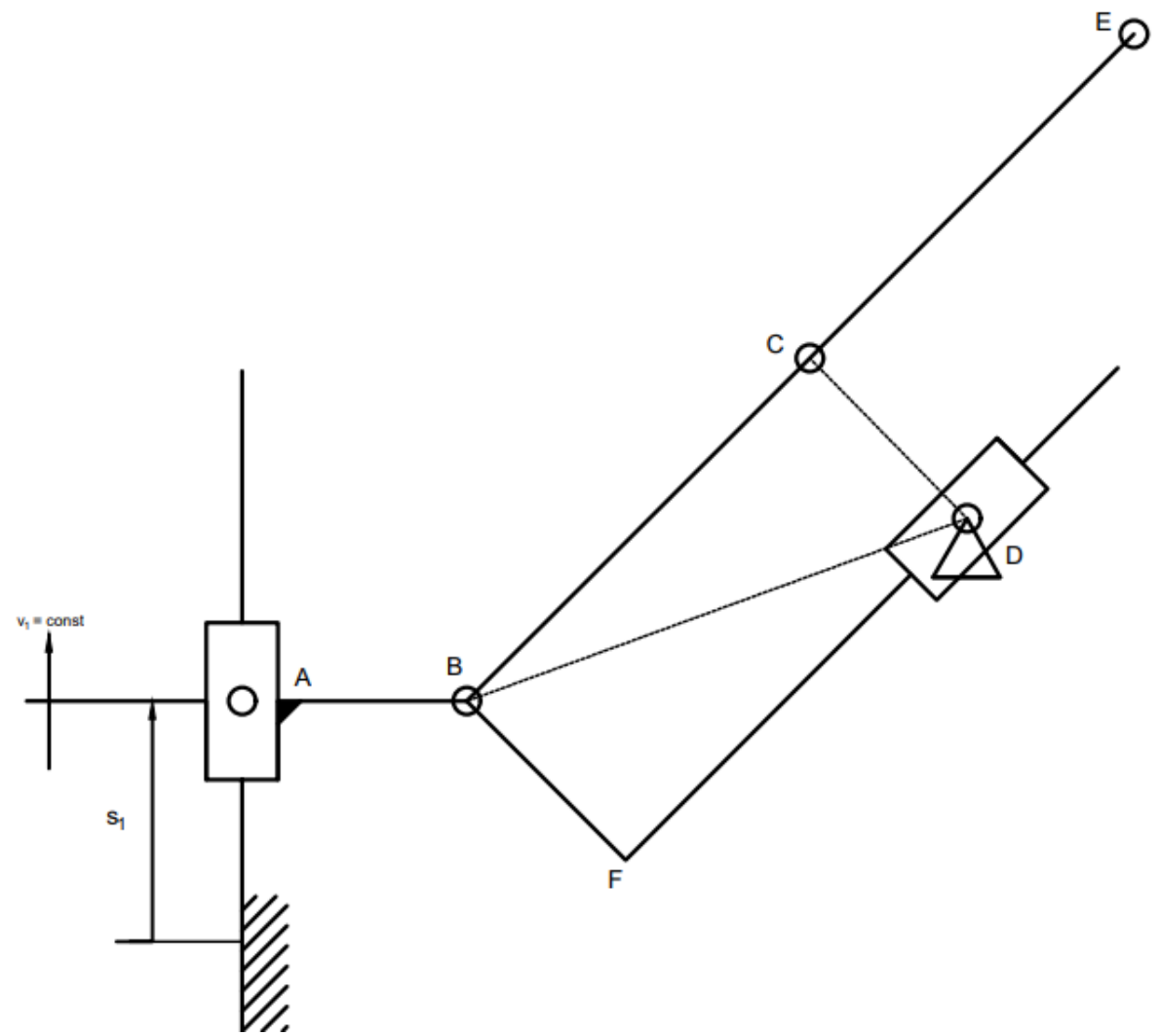

Fig. 3 - Equivalent system with slider projected onto point D.

The angular velocity of the beam FD is naturally the same angular velocity ω_2. To find it, it is necessary to obtain the linear velocity v_{D2B} using a system of equations:

$$\overline{v_{D_2}} = \overline{\underline{\underline{v_B}}} + \overline{\underline{v_{D_2B}}} \text{ and } \overline{v_{D_2}} = \overline{\underline{\underline{v_{D_3}}}} + \overline{\underline{v_{D_2D_3}}}, \text{ therefore:}$$

$$\overline{\underline{\underline{v_B}}} + \overline{\underline{v_{D_2B}}} - \overline{\underline{\underline{v_{D_3}}}} - \overline{\underline{v_{D_2D_3}}} = 0 \Rightarrow \overline{\underline{\underline{v_B}}} + \overline{\underline{v_{D_2B}}} = \overline{\underline{v_{D_2D_3}}},$$

where v_B and v_{D3} (equal 0) are known and the remaining two have known directions, so they can be obtained with the use of a vector polygon (Fig. 4).

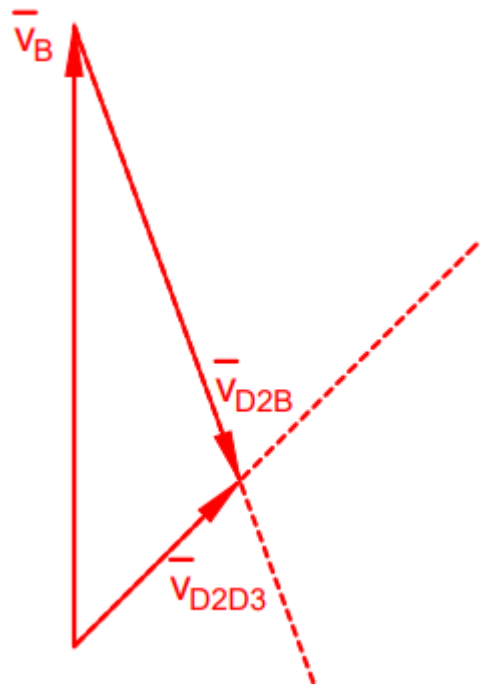

Fig. 4 - Vector polygon for calculating v_{D2B}.

From the figure above, $v_{D_2B} \simeq 0.39\frac{m}{s}$. Now, to calculate the angular velocity:

$$\overline{v_{D_2B}} = \overline{\omega_2} \times \overline{r_{BD}},$$

The result: $\omega_2 = \omega_3 \simeq 0.65\frac{rad}{s}$, clockwise.

Having obtained the angular velocities, the original system will be sufficient for calculating the linear velocities of the points, starting with point C:

$$\overline{v_C} = \overline{\omega_3} \times \overline{r_{CB}} \Rightarrow v_C \simeq 0.16\frac{m}{s}, \text{ upwards along link 2.}$$

To calculate the velocities of points S_2 and E, first v_{S_2B} and v_{EB} are needed:

$$\overline{v_{S_2B}} = \overline{\omega_2} \times \overline{r_{S_2B}} \simeq 0.21\frac{m}{s}, \perp BS_2;$$

$$\overline{v_{EB}} = \overline{\omega_2} \times \overline{r_{EB}} \simeq 0.68\frac{m}{s}, \perp BE.$$

Now, to construct vector polygons for the equations:

$$\overline{v_{S_2}} = \underline{\underline{\overline{v_B}}} + \underline{\underline{\overline{v_{S_2B}}}} \text{ and } \overline{v_E} = \underline{\underline{\overline{v_B}}} + \underline{\underline{\overline{v_{EB}}}}.$$

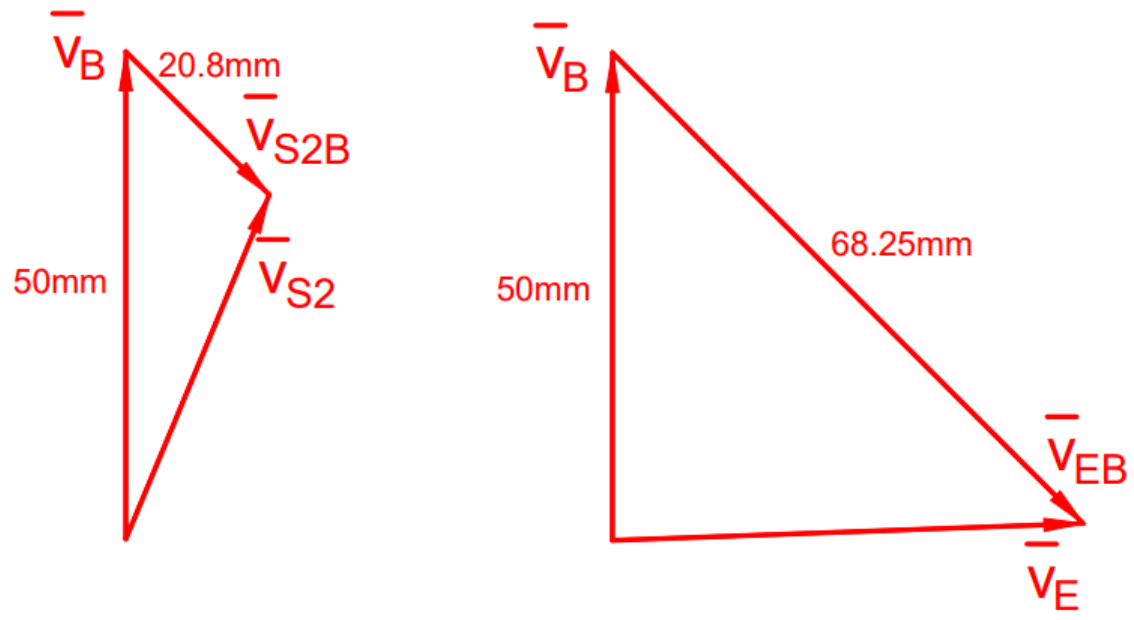

Fig. 5 - Vector polygons for calculating v_{S2} and v_E.

The results:

$$v_{S_2} = 0.38\frac{m}{s} \text{ and } v_E = 0.48\frac{m}{s}.$$

Since points A and B have constant velocities, their accelerations are 0.

To calculate the accelerations of the points, first the angular acceleration of link 2 has to be found. Similarly to angular velocity, it is necessary to use the equivalent system (Fig. 3). From it:

$$\overline{a_{D_2}} = \overline{\underline{\underline{a_B}}} + \overline{\underline{\underline{a^{-n}_{D_2B}}}} + \overline{\underline{a^{-t}_{D_2B}}} \text{ and } \overline{a_{D_2}} = \overline{\underline{\underline{a_{D_3}}}} + \overline{\underline{a^{-t}_{D_2D_3}}} + \overline{\underline{\underline{a^{-cor}}}}, \text{ so:}$$

$$\overline{\underline{a^{-t}_{D_2B}}} = \overline{\underline{a^{-t}_{D_2D_3}}} + \overline{\underline{\underline{a^{-cor}}}} - \overline{\underline{\underline{a^{-n}_{D_2B}}}}.$$

To calculate $a^{-t}_{D_2B} = 0.39\frac{m}{s^2}$, a vector polygon was made (Fig. 6) and from it:

$$\overline{\varepsilon}_2 = \overline{\varepsilon}_3 = \frac{\overline{a^{-t}_{D_2B}}}{\overline{r_{DB}}} \simeq 0.65\frac{rad}{s^2}, \text{ counterclockwise.}$$

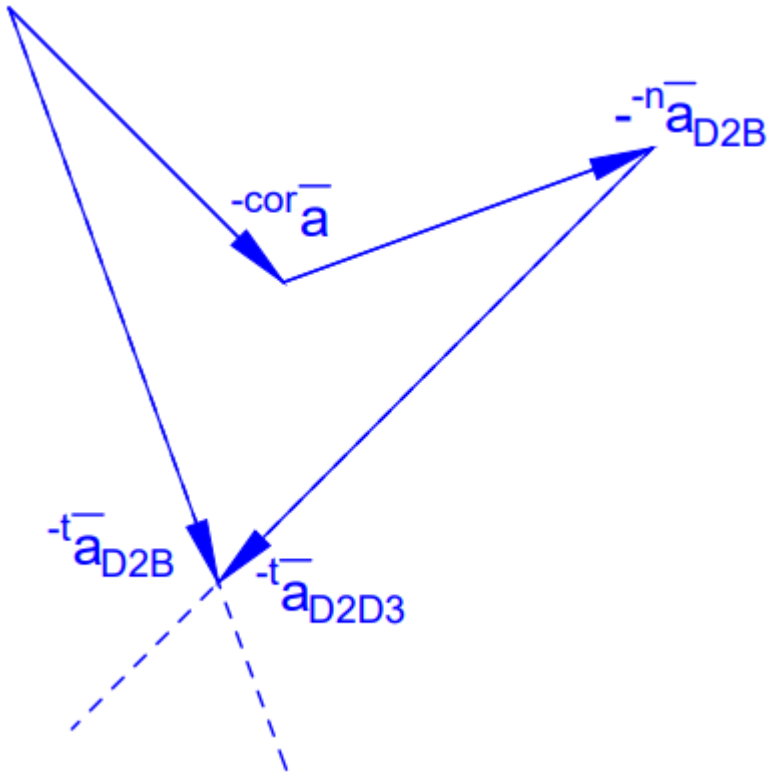

Fig. 6 - Vector polygon for calculating a_{D2B}^{-t}.

Having obtained the angular accelerations, now the linear accelerations of the points are obtainable from the main model.

$$\overline{a_C} = \overline{\underline{\underline{a_{CD}^{-n}}}} + \overline{\underline{\underline{a_{CD}^{-t}}}};$$

$$\overline{a_{S_2}} = \overline{\underline{\underline{a_{S_2B}^{-n}}}} + \overline{\underline{\underline{a_{S_2B}^{-t}}}};$$

$$\overline{a_E} = \overline{\underline{\underline{a_{EB}^{-n}}}} + \overline{\underline{\underline{a_{EB}^{-t}}}}.$$

The results (from Fig. 7):

$$a_C = 0.15\frac{m}{s^2},\ a_{S_2} = 0.25\frac{m}{s^2},\ a_E = 0.59\frac{m}{s^2}.$$

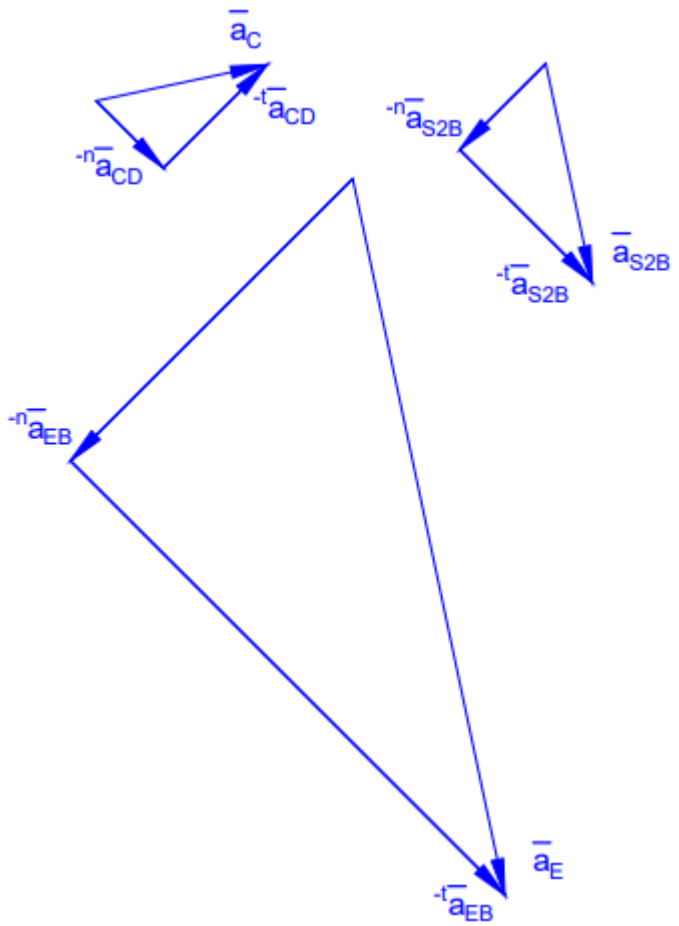

Fig. 7 - Vector polygons for calculating the accelerations.

Analytical approach

Assumptions:

s_1 = 250mm, l_1 = 250mm, l_3 = 250mm, l_4 = 455mm , l_5 = 809mm, φ_3 = 315°.

Once the vector loop is built (fig. 6), it is necessary to calculate initial values of the non constant vectors l_2 and φ_2:

$$\overline{s_1} + \overline{I_1} + \overline{I_2} + \overline{I_3} + \overline{I_4} + \overline{I_5} = 0.$$

The equation above can be broken down for the x and y axes:

- x: $I_1 + I_2 cos\varphi_2 + I_3 cos\,\varphi_3 - I_5 = 0;$
- y: $s_1 + I_2 sin\,\varphi_2 + I_3 sin\,\varphi_3 - I_4 = 0.$

Since there are two unknowns for two equations. Substituting l_2 from the first equation into the second:

$$s_1 + \frac{I_5 - I_1 - I_3 cos\,\varphi_3}{cos\varphi_2} sin\,\varphi_2 + I_3 sin\,\varphi_3 - I_4 = 0 \Rightarrow tg\varphi_2 = \frac{I_4 - s_1 - I_3 sin\,\varphi_3}{I_5 - I_1 - I_3 cos\,\varphi_3} \Rightarrow \varphi_2 = 45°,$$

$$I_2 = \frac{I_5 - I_1 - I_3 cos\,\varphi_3}{cos\varphi_2} = 540\,mm.$$

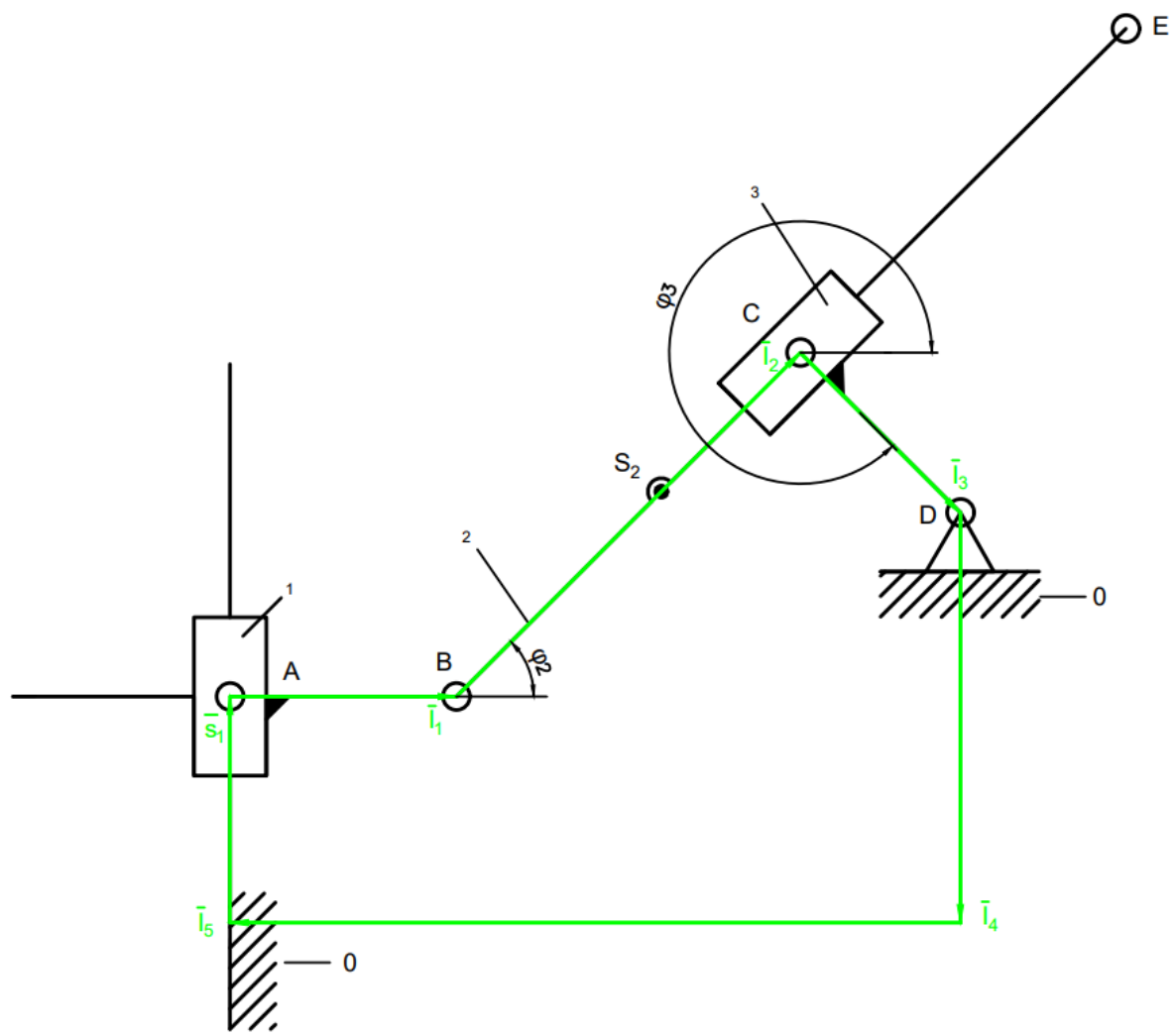

Fig. 8 - The vector loop.

Now, the two equations need to be differentiated (with v_1, ω_2 and ω_3 substituted in):

- x: $\frac{dI_2}{dt} cos\varphi_2 - I_2\omega_2 sin\varphi_2 - I_3\omega_3 sin\varphi_3 = 0;$
- y: $v_1 + \frac{dI_2}{dt} sin\varphi_2 + I_2\omega_2 cos\varphi_2 + I_3\omega_3 cos\,\varphi_3 = 0.$

First, the angular velocities ($\omega_2 = \omega_3$) need to be calculated:

- x: $\frac{dI_2}{dt} = \frac{I_2\omega_2 sin\varphi_2 + I_3\omega_3 sin\varphi_3}{cos\varphi_2} = \omega_2 \frac{I_2 sin\varphi_2 + I_3 sin\varphi_3}{cos\varphi_2};$
- y: $\frac{dI_2}{dt} = \frac{-v_1 - I_2\omega_2 cos\varphi_2 - I_3\omega_3 cos\,\varphi_3}{sin\varphi_2}.$

Then, after substituting ω_2 for ω_3, since they are equal:

$$\omega_2 sin\varphi_2(I_2 sin\varphi_2 + I_3 sin\varphi_3) = cos\varphi_2(-v_1 - I_2\omega_2 cos\varphi_2 - I_3\omega_2 cos\,\varphi_3)$$

$$\omega_2 sin\varphi_2(I_2 sin\varphi_2 + I_3 sin\varphi_3) = -v_1 cos\varphi_2 - I_2\omega_2 cos^2\varphi_2 - I_3\omega_2 cos\varphi_2 cos\,\varphi_3$$

$$\omega_2[sin\varphi_2(I_2 sin\varphi_2 + I_3 sin\varphi_3) + I_2 cos^2\varphi_2 - I_3 cos\varphi_2 cos\,\varphi_3] = -v_1 cos\varphi_2$$

Since φ_3 = 315° = 360° - φ_2 ⇒ cosφ_3 = cosφ_2 and sinφ_3 = -sinφ_2. Substituting:

$$\omega_2 = \frac{-v_1 cos\varphi_2}{I_2 sin^2\varphi_2 + I_3 sin\varphi_2 sin\varphi_3 + I_2 cos^2\varphi_2 - I_3 cos\varphi_2 cos\,\varphi_3} = \frac{-v_1 cos\varphi_2}{I_2} = -0.65\frac{rad}{s}.$$

Knowing $\omega_2 = \omega_3$ is the same as in the graphoanalytical approach and all lengths are the same, it can be concluded that the linear velocities of the points will be equal as well.

To calculate the accelerations, the two equations need to be differentiated again:

- x: $\frac{d^2I_2}{dt^2}cos\varphi_2 - \frac{dI_2}{dt}\omega_2 sin\varphi_2 - \frac{dI_2}{dt}\omega_2 sin\varphi_2 - I_2\varepsilon_2 sin\varphi_2 - I_2\omega_2^2 cos\varphi_2 - I_3\varepsilon_3 sin\varphi_3 - I_3\omega_3^2 cos\varphi_3 = 0;$
- y: $\frac{d^2I_2}{dt^2}sin\varphi_2 + \frac{dI_2}{dt}\omega_2 cos\varphi_2 + \frac{dI_2}{dt}\omega_2 cos\varphi_2 + I_2\varepsilon_2 cos\varphi_2 - I_2\omega_2^2 sin\varphi_2 + I_3\varepsilon_3 cos\varphi_3 - I_3\omega_3^2 sin\varphi_3 = 0.$

Again, substituting $\omega_2 = \omega_3$, $\varepsilon_2 = \varepsilon_3$, cos$\varphi_3$ = cosφ_2 and sinφ_3 = -sinφ_2:

- x: $\frac{d^2I_2}{dt^2} = \frac{2\frac{dI_2}{dt}\omega_2 sin\varphi_2 + I_2\varepsilon_2 sin\varphi_2 + I_2\omega_2^2 cos\varphi_2 - I_3\varepsilon_2 sin\varphi_2 + I_3\omega_2^2 cos\varphi_2}{cos\varphi_2};$
- y: $\frac{d^2I_2}{dt^2} = \frac{-2\frac{dI_2}{dt}\omega_2 cos\varphi_2 - I_2\varepsilon_2 cos\varphi_2 + I_2\omega_2^2 sin\varphi_2 - I_3\varepsilon_2 cos\varphi_2 - I_3\omega_2^2 sin\varphi_2}{sin\varphi_2}.$

$$2\frac{dI_2}{dt}\omega_2 sin^2\varphi_2 + I_2\varepsilon_2 sin^2\varphi_2 + I_2\omega_2^2 sin\varphi_2 cos\varphi_2 - I_3\varepsilon_2 sin^2\varphi_2 + I_3\omega_2^2 sin\varphi_2 cos\varphi_2 =$$
$$= -2\frac{dI_2}{dt}\omega_2 cos^2\varphi_2 - I_2\varepsilon_2 cos^2\varphi_2 + I_2\omega_2^2 sin\varphi_2 cos\varphi_2 - I_3\varepsilon_2 cos^2\varphi_2 - I_3\omega_2^2 sin\varphi_2 cos\varphi_2$$

$$2\frac{dI_2}{dt}\omega_2 + I_2\varepsilon_2 + 2I_3\omega_2^2 sin\varphi_2 cos\varphi_2 = 0$$

$$\varepsilon_2 = \varepsilon_3 = \frac{-2\frac{dI_2}{dt}\omega_2 - 2I_3\omega_2^2 sin\varphi_2 cos\varphi_2}{I_2} = -0.65\frac{rad}{s^2}.$$

Having obtained the angular accelerations equal to the values for the graphoanalytical analysis, the linear accelerations will also be the same for both methods.

Velocities and accelerations	Graphoanalytical	Analytical
v_B, a_B	0.5 $\frac{m}{s}$, 0$\frac{m}{s^2}$	0.5 $\frac{m}{s}$, 0$\frac{m}{s^2}$
v_C, a_C	0.16 $\frac{m}{s}$, 0.11$\frac{m}{s^2}$	0.16 $\frac{m}{s}$, 0.11$\frac{m}{s^2}$
v_E, a_E	0.48 $\frac{m}{s}$, 0.59$\frac{m}{s^2}$	0.48 $\frac{m}{s}$, 0.59$\frac{m}{s^2}$
v_{S2}, a_{S2}	0.38 $\frac{m}{s}$, 0.25$\frac{m}{s^2}$	0.38 $\frac{m}{s}$, 0.25$\frac{m}{s^2}$
$\omega_2 = \omega_3$, $\varepsilon_2 = \varepsilon_3$	0.65 $\frac{rad}{s}$, 0.65$\frac{rad}{s^2}$	0.65 $\frac{rad}{s}$, 0.65$\frac{rad}{s^2}$

Table. 1 - Comparison of values for different methods - results exactly the same.

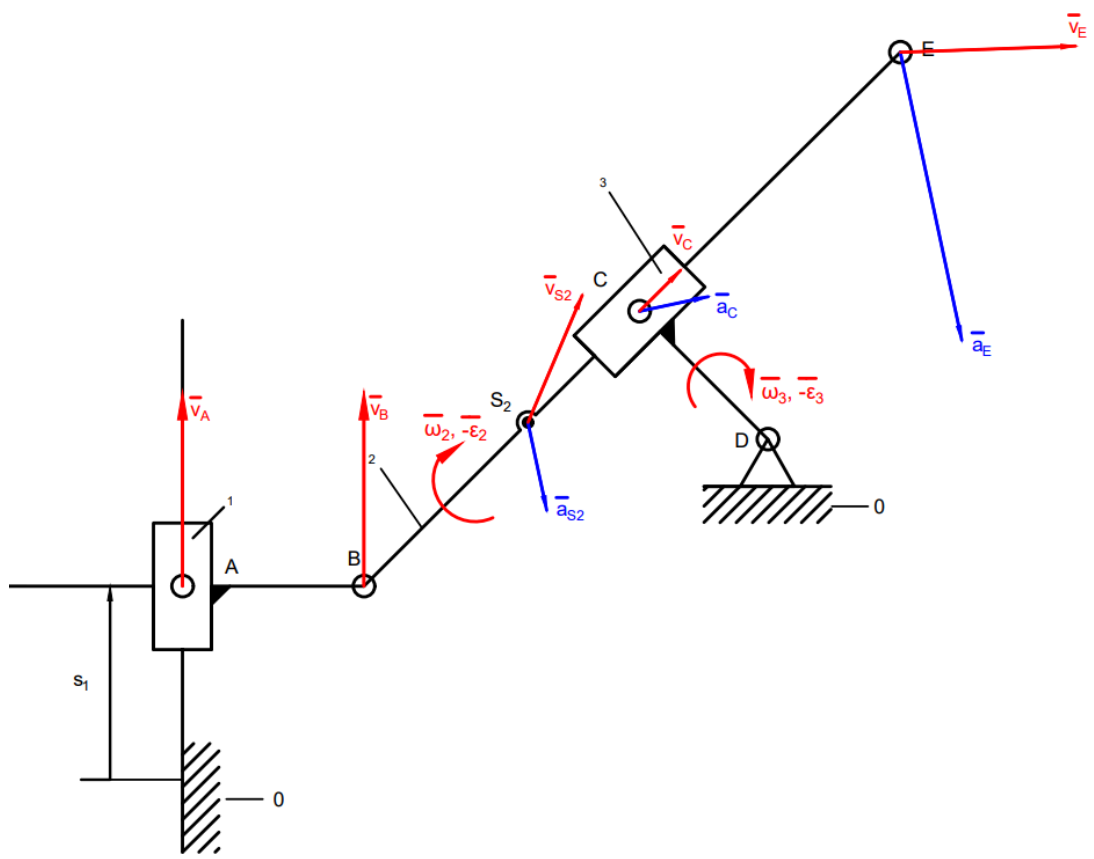

Fig. 9 - The model after kinematic analysis.

3. Dynamic analysis

Graphoanalytical approach

Assumptions:

$m_2 = 1kg, \quad J_{S_2} = 10\frac{kg}{m^2}, \quad P_2 = 20N, \quad M_3 = 10Nm,$ scale: $10mm = 1N.$

Forces: $G = m_2 g = 10N, \quad B_2 = m_2 a_{S_2} = 0.25N, \quad M_2 = J_{S_2}\varepsilon_2 = 6.55Nm.$

From Fig. 8: h_1 = 150mm, h_2 = 200mm, h_3 = 360mm.

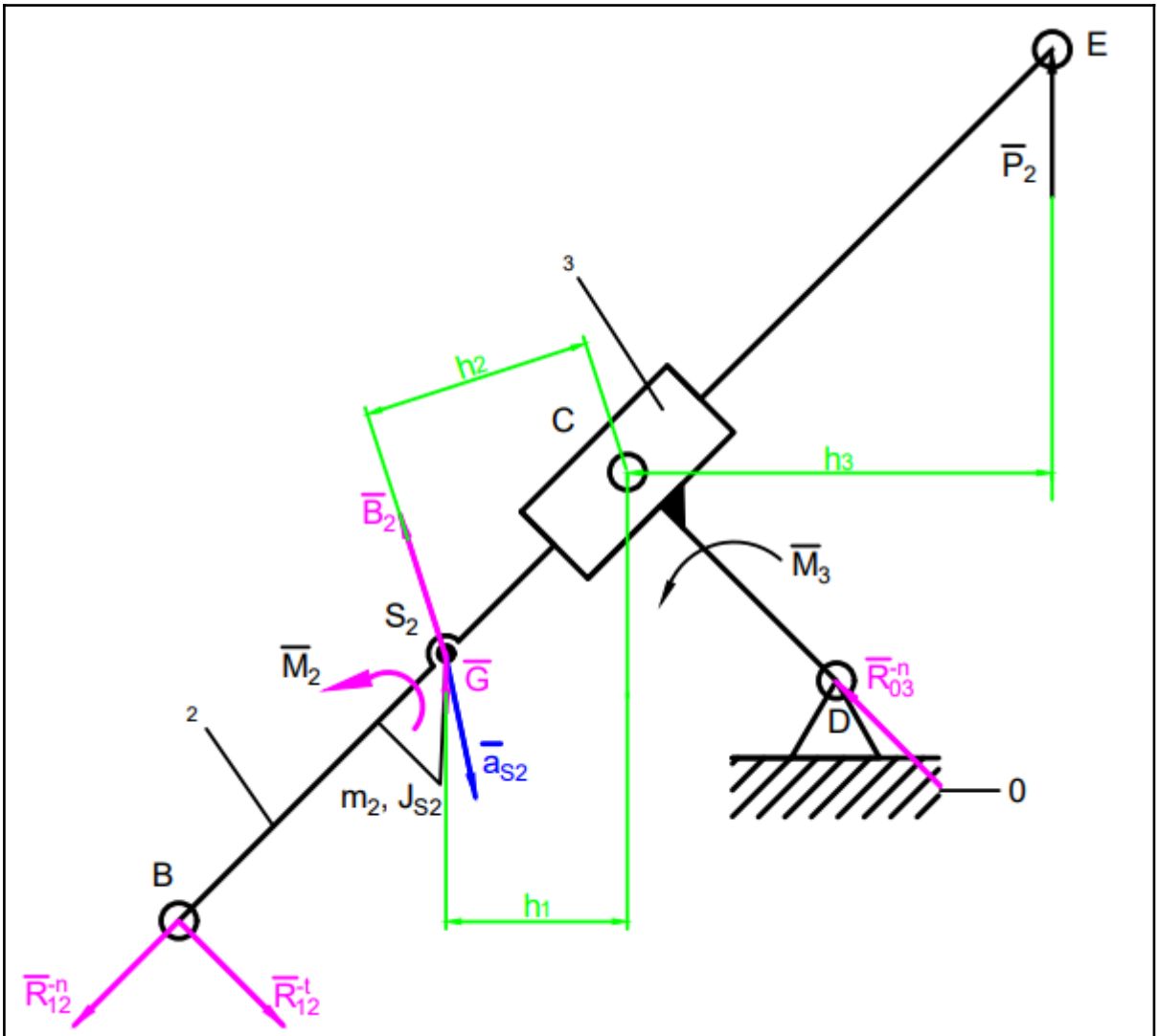

Fig. 10 - The model after releasing the group (2,3) from boundaries. [not yet in scale]

The vector equilibrium equation:

$$\sum \overline{P}_{i(2,3)} = \underline{\overline{R_{12}^{-n}}} + \underline{\overline{R_{12}^{-t}}} + \underline{\underline{\overline{B_2}}} + \underline{\underline{\overline{G}}} + \underline{\overline{R_{03}^{-n}}} + \underline{\underline{\overline{P_2}}} = 0$$

There are three values of unknown magnitudes, so a moment equation about point C will be necessary:

$$\sum M_{iC(2,3)} = R_{12}^{-t} \cdot BC + G \cdot h_1 - B_2 \cdot h_2 + M_3 + P_2 \cdot h_3 + M_2 = 0$$

From it, the tangential reaction at point B can be obtained:

$$R_{12}^{-t} = \frac{-G \cdot h_1 + B_2 \cdot h_2 - M_3 - P_2 \cdot h_3 - M_2}{BC} = -46.67N.$$

As a result, the force equilibrium equation takes form:

$$\sum \overline{P}_{i(2,3)} = \underline{\overline{R_{12}^{-n}}} + \underline{\underline{\overline{R_{12}^{-t}}}} + \underline{\underline{\overline{B_2}}} + \underline{\underline{\overline{G}}} + \underline{\overline{R_{03}^{-n}}} + \underline{\underline{\overline{P_2}}} = 0.$$

Now, to calculate the two remaining forces, the vector polygon needs to be built.

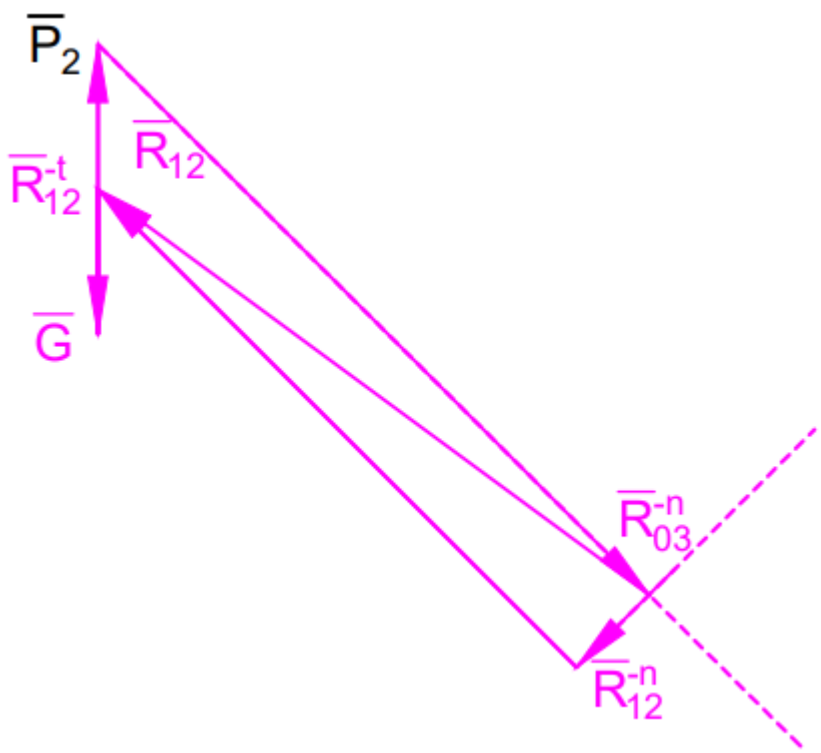

Fig. 11 - Vector polygon for calculating the reaction forces (B_2 neglected).

From the figure: $R_{12}^{-n} = 7.07N$, $R_{03}^{-n} = R_{03} = 53.74N$, as well as $R_{12} = 47.2N$.

The reaction force R_{12} can be used to determine the generalised external load (Fig. 12).

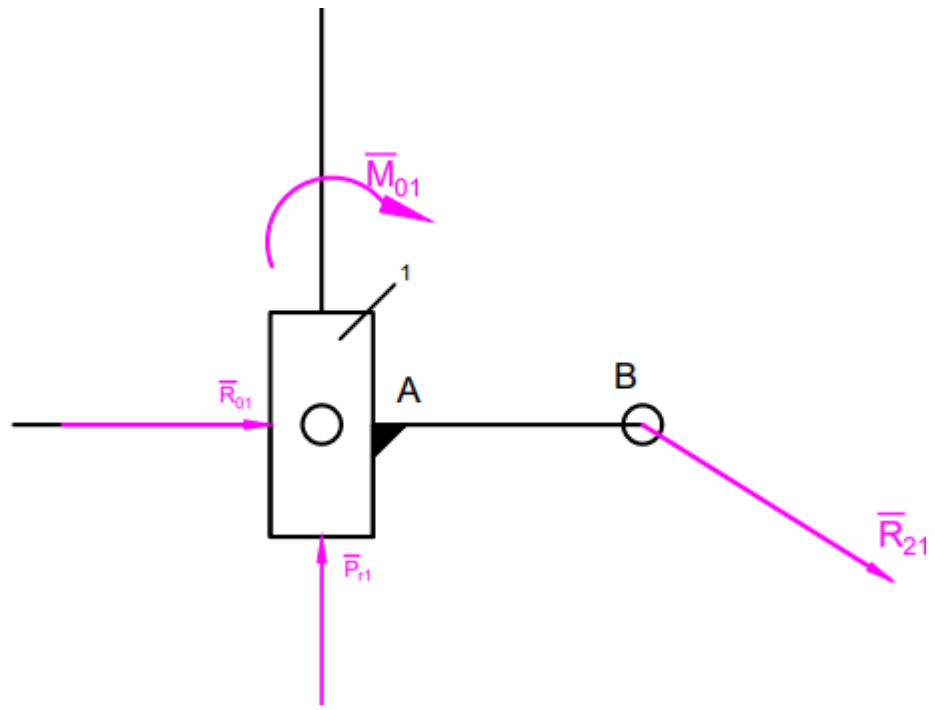

Fig. 12 - Link A with generalised load.

The angle between R_{21} and the horizontal is 36°, so:

$$R_{21} cos\, 36° = R_{01} = 38.19N \text{ and } R_{21} sin\, 36° = P_{r1} = 27.74N.$$

Also, the moment M_{01}:

$$M_{01} = P_{r1} \cdot AB = 6.94Nm.$$

Analytical approach

For the analytical approach, the same initial assumptions as before are considered. Each link has to be drawn separately with its corresponding forces.

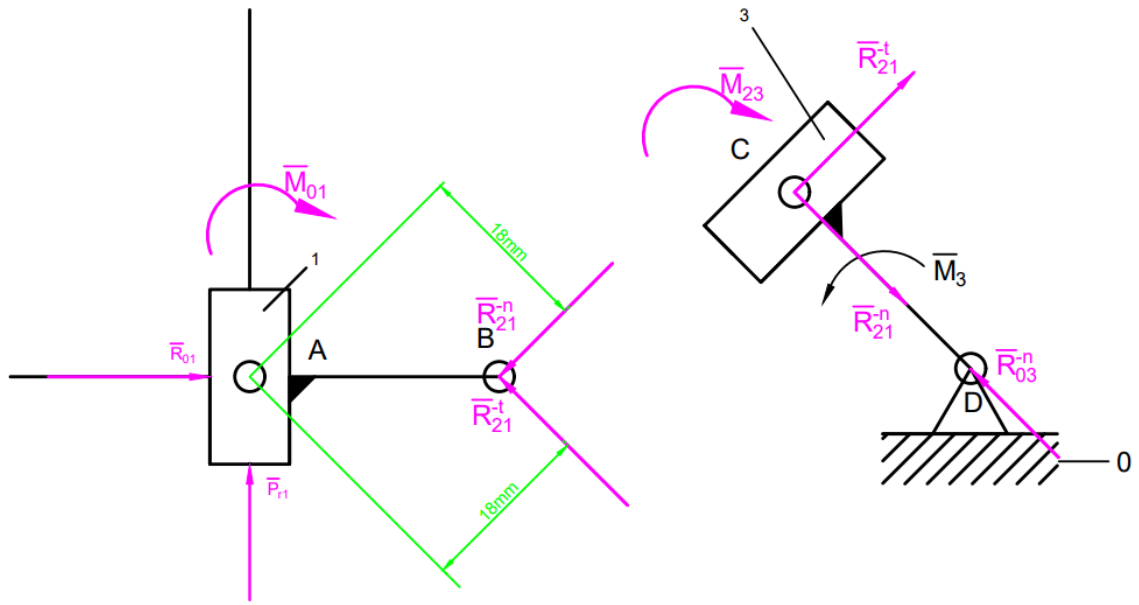

Fig. 13 - Links A and C.

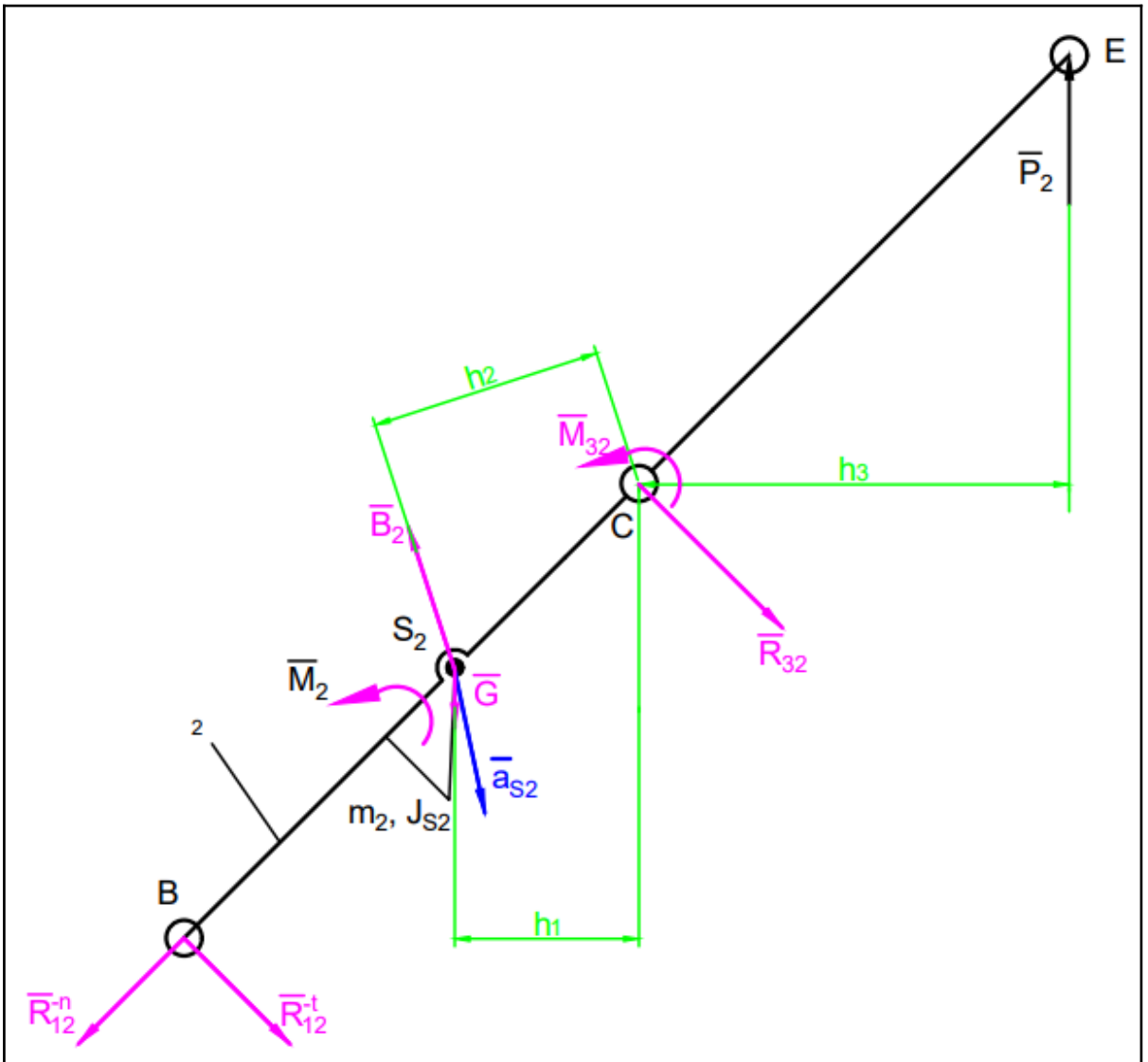

Fig. 14 - Link B.

Starting with link 1:

- x: $R_{01} - R_{21}^{-n} cos\, 45° - R_{21}^{-t} sin\, 45° = 0;$
- y: $P_{r1} - R_{21}^{-n} sin\, 45° - R_{21}^{-t} cos\, 45° = 0;$
- M_{iB}: $P_{r1} \cdot 0.25 - M_{21} = 0;$

Link 2:

- x: $R_{12}^{-n} cos\, 45° + R_{12}^{-t} cos\, 45° - B_2 sin\, 12° + R_{32} cos\, 45° = 0;$
- y: $R_{12}^{-n} sin\, 45° - R_{12}^{-t} sin\, 45° + B_2 cos\, 12° - G - R_{32} sin\, 45° + P_2 = 0;$
- M_{iC}: $R_{12}^{-t} \cdot BC + M_2 - B_2 \cdot h_2 + G \cdot h_1 + P_2 \cdot h_3 = 0;$

Link 3:

- x: $-R_{23}^{-n}cos\,45° - R_{23}^{-t}cos\,45° - R_{03}cos\,45° = 0;$
- y: $R_{23}^{-n}sin\,45° - R_{23}^{-t}sin\,45° + R_{03}sin\,45° = 0;$
- M_{iC}: $M_3 - M_{23} = 0;$

Starting from link 2, similarly to the graphoanalytical approach, R_{12}^{-t} can be obtained from the moment equation: $R_{12}^{-t} =- 46.67N$.

Having it, the equations for x and y can be subtracted to give:

$$R_{12}^{-t}cos\,45° - B_2(sin\,12° + cos\,12°) + G - P_2 + 2R_{32}cos\,45° = 0,$$

With only R_{32} as the unknown: $R_{32} = 39.67N$.

Substituting back into x: $R_{12}^{-n} = 7.07N$.

With both R_{12}^{-t} and R_{12}^{-n}, it is possible to get $R_{12} = 47.2N$, which, substituted into the link 1 equations will give: $R_{01} = 38.19N$, $P_{r1} = 27.74N$ and $M_{01} = 6.94N$.

Similarly, R_{12} can be substituted into link 3's equations to obtain $R_{03} = 53.74N$.

Forces and moments	Graphoanalytical	Analytical
P_{r1}	27.74N	27.74N
R_{01}	38.19N	38.19N
M_{01}	6.94Nm	6.94Nm
R_{12}	47.2N	47.2N
R_{03}	53.74N	53.74N

Table. 2 - Comparison of values for different methods - results exactly the same.

Method of instantaneous powers

Finally, to confirm the results, the method of instantaneous powers can be used. As the sum of all powers in the system should be 0, multiplying forces by their corresponding velocities should go to 0.

$$P_{r1}v_A + B_2v_{S_2}cos\,35° + Gv_{S_2}cos\,157° - M_2\omega_2 - M_3\omega_3 + P_2cos\,88° = 0$$

From it, P_{r1} = 26.92N, within 1N from the analytically obtained value, possibly due to rounding.